Midterm

CSED501

Taylor Kessler Faulkner

Due: **November 11, 2025**

Name: Aeden Jameson

Instructions: Complete the questions below and upload a PDF containing your written answers to Gradescope.

- Keep your answers within the provided space.
- Answers can be handwritten, or typed and pasted into this PDF.
- Show your work; this makes giving partial credit easier!
- Numeric answers can be given in fraction or decimal form; if decimal, round to 2 places

Problem 1 8 pts**:**

True/False. Mark each statement as True or False and briefly justify your answer.

(a) (2 pts) The variance of a model generally **decreases** as the number of features **increases**.

(b) (2 pts) The variance of a model generally **decreases** as the weight of the regularization error **increases**.

(c) (2 pts) Assume we train a model on a given dataset. If we were to remove 50% of samples from the dataset and re-train the model from scratch, the new model will be more likely to overfit to its training data than the old one.

(d) (2 pts) A solution to a convex optimization problem is guaranteed to be a global minimum.

Answer:

a) False. Typically the variance increase as features increase because the model is more flexible, which means the learned f can vary widely and different training sets.

b) True. Regularization shrinks the parameters that are learned, in the case of L1 can set those values to 0, and so the learned model doesn't overfit and consequently vary widely if trained on different training sets.

c) True. In practice probably because good datasets are hard to collect and the full set probably still has issues. As a result of less data the model doesn't generalize. However, theoretically if that 50% had enough signal then no it wouldn't necessarily overfit.

d) True. One of the mathmatical properties of a convex optimzation problem is that every local minimum is a global minimum so if a solution exists that solution is a global minimum.

Problem 2 $\boxed{\textbf{24 pts}}$**:**

Short Answer. For each problem, provide a short answer and explanation(1 paragraph). When providing ML methods, only provide ML methods that we have covered in class.

(a) (8 pts) You have an unlabeled data set containing descriptions of two types of products, Product A and Product B. This description includes features such as color, weight, and other relevant information. You are interested in grouping the data to try to separate out Product A and Product B, but you don't have time to label the dataset by hand. What ML method(s) would you try, and why?

(b) (8 pts) You work for a large clothing company, and your boss wants you to write a program that predicts whether a new item of clothing will sell well based on features such as price, material, and color. Your boss wants you to train a model fairly quickly, but is willing to purchase the compute needed to parallelize code. What ML method(s) would you try, and why?

(c) (8 pts) Your coworker has been recording how long it takes them to finish coding tasks at work for the last year, and has noted features of each task such as the number of files that need to be written or edited, the number of people working on the task, the perceived difficulty of the task, and others. They ask you to write a program to predict how long a coding task will take them to finish. What ML method(s) would you try, and why?

Answer:

a) Given that I don't have labels for the dataset I'd start by using unsupervised methods. I'd begin with K-means (k=2) for it's simpilicity. If the coverage of the clusters was off I would also try a Gaussian Mixture model. Additionally, I would try the clustering algorithms in combination with PCA.

b) The problem statement is essentially asking whether an item will sell or not based on n features. As a baseline I would identify the accuracy of the majority class classifier. From there I would use both parametric and non-parametrics methods: specifically logistic regression with L2 and/or L1 and a random forest in order to cover the case that the data may not be linearly seperable. In addition both can be parallelized. If I have time to spare, I could try SVM on the side.

c) I would use both parametric and non-parametrics methods to predict the length of a coding task. That allows me to choose the best model for the traits of the dataset. e.g. imbalanced, outliers, not linearly separable

* I'd start with linear regression as a baseline and if that didn't do well enough L2 and/or L1 regularization to deal with potential multi-colinearity.

* In addition I'd try a random first for it's ability to deal with non-linear factors that contriubte like number of reviewers and perceived complexity of the task. RF's can also deal the how features interact. e.g. experience with the code base and difficulty of the task.

Problem 3 $\boxed{\textbf{28 pts}}$**:**

In Figure 1 We have two labeled datasets with two and three classes, respectively. Assume that within each dataset the classes are separable, albeit not necessarily linearly separable. Furthermore, assume that no two distances are equal and that we implement the algorithms below on a magical computer with no rounding error.

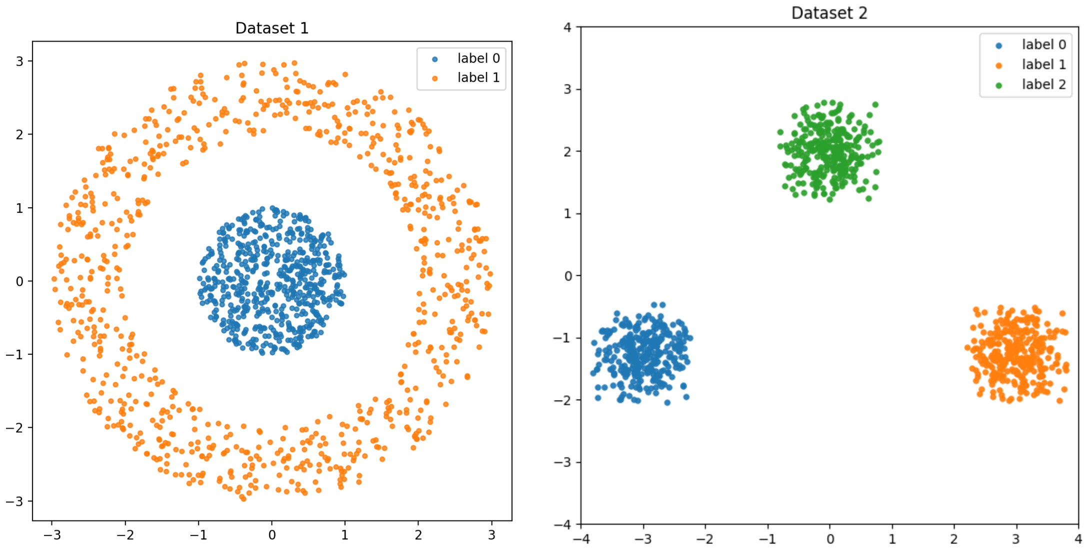

Figure 1: The two datasets. Colors correspond to labels.

(a) (2 pts) Which dataset is linearly separable and which one is not?

(b) (5 pts) For the dataset that is linearly separable, how many weight vectors would **softmax logistic regression** learn, and what are the dimensions of those weight vectors?

(c) (5 pts) For the dataset that is not linearly separable, describe a feature transformation that makes the data linearly separable in that feature space.

(d) (8 pts) For the assumptions in the intro, and $k = 2$ on the first dataset and $k = 3$ on the second dataset, is Agglomerative Clustering using single linkage **guaranteed** to correctly identify the clusters in each dataset (that is, are you guaranteed to be able to cut off Agglomerative Clustering using single linkage at a point that identifies the clusters shown)? Briefly explain why/why not.

(e) (8 pts) For the assumptions in the intro, and $k = 2$ on the first dataset and $k = 3$ on the second dataset, is K-means **guaranteed** to correctly identify the clusters in each dataset? Briefly explain why/why not.

Answer:

a) The dataset on the left is not linearly separable because no single straight line can separate the rings. The dataset with the three blobs on the right is linearly separable in that one can use a one vs all strategy in which a straight line can be drawn between each pair of two classes.

b) Since there are k=3 class labels then 3 weight vectors with length two would be learned. Those dimensions result in the following defintion of softmax in our problem,

$$p(y=k|x) = \text{softmax}\,(Wx+b),\ \ W \in R3{\times}2,\, b \in R3$$

c) A feature transformation that would make the first dataset linearly separable is a degree two polynomial or RBF.

d) For the dataset on the left yes because all the edges for the points in inner circle are close together and same is true for all the points in the outer circle of the dendrogram. There will be only one long edge for getting from the inner circle to the outter ring. For the dataset on the right, the algorithm should also work because a visual spot check reasonably concludes that the two point farthest points away within the groups is shorter than the distance between the two closest points between the groups.

e) When using k-means for the dataset on the left the clusters will not be correctly identified because the algorithm assumes linearly separable groups inside a circle. We've established linear separability is not possible for this dataset.

For the dataset on the right, probably because the groups are well separated and can surronded by a circle in theory.

Problem 4 $\boxed{\textbf{18 pts}}$**:**

In this problem, we will show that the Perceptron algorithm can be interpreted as performing stochastic gradient descent with a batch size of 1 on a hinge-like loss. Recall that the Perceptron algorithm classifies examples (x_i, y_i) with $y_i \in \{-1, +1\}$ using a linear model $f(x; w) = w^\top x$ and the classification rule:

$$\hat{y}_i = \begin{cases} 1, & \text{if } f(x_i; w) \geq 0, \\ -1, & \text{if } f(x_i; w) < 0. \end{cases}$$

(a) (5 pts) Write the Perceptron update rule, in vector form, for a point (x_i, y_i) and step size η. Write your answer in the following format:

$$w^{t+1} \leftarrow \begin{cases} \text{TODO}, & \text{if } y_i = \hat{y}_i, \\ \text{TODO}, & \text{if } y_i \neq \hat{y}_i. \end{cases}$$

(b) (8 pts) Define a *hinge-like* loss function

$$\ell(w; x_i, y_i) = \max(0,\ -y_i \cdot f(x_i; w)).$$

We will compute the gradient $\nabla_w \ell(w; x_i, y_i)$. Note that the function $m(a) = \max(0, a)$ is not differentiable at $a = 0$. For simplicity, assume that

$$\frac{\delta m(a)}{\delta a} = \begin{cases} 0, & \text{if } a < 0, \\ 1, & \text{if } a \geq 0. \end{cases}$$

Let $a(w; x_i, y_i) = -y_i \cdot f(x_i; w)$. Using the chain rule,

$$\nabla_w \ell(w; x_i, y_i) = \frac{\delta m(a(w; x_i, y_i))}{\delta a} \cdot \nabla_w a(w; x_i, y_i).$$

Complete the derivation of $\nabla_w \ell(w; x_i, y_i)$ in the following case format (reminder: $\nabla_w a(w; x_i, y_i)$ is the gradient of $a(w; x_i, y_i)$ with respect to w.

$$\nabla_w \ell(w; x_i, y_i) = \begin{cases} \text{TODO}, & \text{if } a < 0, \\ \text{TODO}, & \text{if } a \geq 0. \end{cases}$$

(c) (5 pts) Show that the Perceptron update from part (a) is equivalent to performing one stochastic gradient descent step for the point (x_i, y_i) with the same step size η:

$$w^{t+1} \leftarrow w^t - \eta \, \nabla_w \ell(w; x_i, y_i).$$

You should again express this as a two-case update as in part (a) to show that the updates are the same. Show your work.

(d) (5 pts) **Extra credit**: Explain why the loss above is called "hinge-like" and describe how it differs from the actual SVM hinge loss. What advantage does the SVM hinge loss have over the loss $\ell(w; x_i, y_i)$ we defined?

Answer:

a) $$w^{t+1} \leftarrow \begin{cases} w^t & \text{if } y_i = \hat{y}_i \\ w^t + \eta y_i x_i & \text{if } y_i \neq \hat{y}_i \end{cases}$$

b) Given $a(w; x_i y_i) = -y_i f(x_i; w)$ then

$$\nabla_w a(w; x_i, y_i) = -y_i \nabla_w f(x_i; w)$$

$$= -y_i \nabla_w w^T x_i = -y_i x_i$$

Now using $\frac{\partial m(a)}{\partial a}$ we can write

$$\nabla_w l(w; x_i, y_i) = \begin{cases} 0 \cdot -y_i x_i = 0 & \text{if } a < 0 \\ 1 \cdot -y_i x_i = -y_i x_i & \text{if } a \geq 0 \end{cases}$$

c) So $w^{t+1} = w^t - \eta \nabla l(w^t; x_i; y_i)$

Using the gradient in part b we can write

$$= \begin{cases} w^t - \eta \cdot 0 = w^t, & y_i w^{t T} x_i > 0 \\ w^t - \eta(-y_i x_i) = w^t + \eta y_i x_i, & y_i w^{t T} x_i \leq 0 \end{cases}$$

Problem 5 $\boxed{\textbf{22 pts}}$**:**

Suppose we have a dataset $\{X_1, \ldots, X_n\}$ of size n where each sample is an i.i.d. (independent and identically distributed) random variable $X_i = \theta + \varepsilon_i$ for some fixed θ and random noise ε_i with $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$. That is, each X_i starts with the same fixed θ value, and then adds random noise. We use three very simple estimators of θ to investigate the bias-variance trade off:

- **Estimate θ using only one sample**: $\hat{\theta}_1 = X_1$.
- **Sample mean**: $\hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^n X_i$.
- **An estimator that is clearly wrong(or is it?)**: $\hat{\theta}_3 = 0.5 \ \cdot X_1$.

We evaluate estimators by their mean squared error (MSE) on a fresh i.i.d. sample $X_{\text{new}} = \theta + \varepsilon_{\text{new}}$. Recall that for some estimator $\hat{\theta}$ of θ, we can split the MSE into three terms:

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}\Big[(X_{\text{new}} - \hat{\theta})^2\Big] = \underbrace{\big(\mathbb{E}[\hat{\theta}] - \theta\big)^2}_{\text{Bias}^2} + \underbrace{\mathrm{Var}(\hat{\theta})}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible Error}}.$$

(a) (4 pts) In your own words, what are the expectations in this formula taken over? That is, what is the source of randomness here?

(b) (9 pts) For $\hat{\theta}_1$, compute the bias squared, variance, and MSE. Your final answers should only be in terms of σ and θ. Show your work. Note: the variance of a fixed value is 0, and $\mathbb{E}[A + B] = \mathbb{E}[A] + \mathbb{E}[B]$, and if A and B are independent, $\mathrm{Var}(A + B) = \mathrm{Var}(A) + \mathrm{Var}(B)$.

- **The Bias^2 term**: $\mathrm{Bias}(\hat{\theta}_i)^2 = (\mathbb{E}[\hat{\theta}_i] - \theta)^2$.
- **The Variance term**: $\mathrm{Var}(\hat{\theta}_i)$.
- **The MSE**: $\mathrm{MSE}(\hat{\theta}_i)$.

(c) (9 pts) We will compute the bias squared, variance, and MSE for $\hat{\theta}_2$ and $\hat{\theta}_3$ for you:

For $\hat{\theta}_2$:

- **The Bias^2 term**:
$\mathrm{Bias}(\hat{\theta}_2)^2 = (\mathbb{E}[\hat{\theta}_2] - \theta)^2 = (\mathbb{E}[X_1] - \theta)^2 = (\mathbb{E}[\frac{1}{n}\sum_{i=1}^n X_i] - \theta)^2 = (\theta + \frac{1}{n}\mathbb{E}[\sum_{i=1}^n \epsilon_i] - \theta)^2 = (\theta - \theta)^2 = 0$.
- **The Variance term**: $\mathrm{Var}(\hat{\theta}_2) = \mathrm{Var}(\frac{1}{n}\sum_{i=1}^n X_i) = \mathrm{Var}(\frac{1}{n}\sum_{i=1}^n \epsilon_i) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(\epsilon_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$.
- **The MSE**: $\mathrm{MSE}(\hat{\theta}_2) = \frac{\sigma^2}{n} + \sigma^2 = \frac{\sigma^2(n+1)}{n}$.

For $\hat{\theta}_3$:

- **The Bias^2 term**:
$\mathrm{Bias}(\hat{\theta}_3)^2 = (\mathbb{E}[\hat{\theta}_3] - \theta)^2 = (\mathbb{E}[0.5 \cdot X_1] - \theta)^2 = (0.5\mathbb{E}[X_1] - \theta)^2 = (0.5\theta - \theta)^2 = 0.25\theta^2$.
- **The Variance term**: $\mathrm{Var}(\hat{\theta}_3) = 0.5^2\mathrm{Var}(X_1) = 0.25\sigma^2$.
- **The MSE**: $\mathrm{MSE}(\hat{\theta}_3) = 0.25\theta^2 + 0.25\sigma^2 + \sigma^2 = 0.25\theta^2 + 1.25\sigma^2$.

QUESTION CONTINUED ON NEXT PAGE

Problem 5 **TODO pts**: CONTINUED FROM PREVIOUS PAGE

Using $\hat{\theta}_1$ as the baseline, briefly answer the following:

- How do $\hat{\theta}_2$ and $\hat{\theta}_3$ compare in terms of bias squared, variance, and overall MSE?
- Which estimator ($\hat{\theta}_2$ or $\hat{\theta}_3$) behaves most like a regularized regression model (e.g., Ridge or Lasso)?
- Is there anything we can do to reduce the irreducible error term?

Answer:

a) The source of randomness is the noise terms in the training samples. That makes the learned estimator random

b) Given $\hat{\theta}_1 = X_1 + \varepsilon_1$ then

$$\text{Bias}^2 = (E[\hat{\theta}_1] - \theta)^2 = (E[X_1] - \theta)^2$$

$$= (E[\theta + \varepsilon_1] - \theta)^2 = (E[\theta] + E[\varepsilon_1] - \theta)^2$$

$$= (\theta + 0 - \theta)^2 = 0$$

$$\text{Var}(\hat{\theta}_1) = \text{Var}(X) = V(\theta + \varepsilon_1) = \sigma^2$$

$\text{var}(\theta) = 0$, θ fixed

$$\text{MSE}(\hat{\theta}_1) = \text{Bias}^2 + \text{Var}(\hat{\theta}_1) + \sigma^2$$

$$= 0 + \sigma^2 + \sigma^2 = 2\sigma^2$$

c.1) The total error terms of theta 1 and 2 compare as follows:
* The biases are both 0
* Variance of theta 2 is much smaller
* The MSE is smaller for n > 1

The total error terms of theta 1 and 3 compare as follows:
* Theta 3 introduces bias
* Theta 3 has lower variance
* Overall MSE of theta 3 is lower than theta 1

c.2) Theta 3 behaves more like ridge and lasso because as we see from the comparion above it trades off bias for variance. Additionally, the co-efficient of 0.5 shrinks every sample and introduces.

c.3) The irreducible error is just that irreducible. It won't change by picking a different estimator. It's the noice in the data from the source that generated the data. To do anything about it one would need to the source of the phenomenom and do more science by controlling more variables or improve the technology that captured the data.